

ATLIQ TECHNOLOGIES

Enterprise AI Delivery — Forward Deployed Engineering

BUSINESS REQUIREMENT DOCUMENT

AI-Powered Field Service Command Centre

Prepared for Vaayu Pumps & Systems Ltd

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ABOUT THIS DOCUMENT — READ BEFORE YOU USE IT

This is a TEACHING ARTEFACT from the Codebasics Forward Deployed AI Engineer Roadmap 2026. Vaayu Pumps & Systems Ltd, AtliQ Technologies and every person named in it are fictional. The numbers are invented but realistic. No real client data appears anywhere in this pack.

It is written the way a real BRD is written, so you can see the shape of the thing before you write your own in Week 19. Two things have been added that a real BRD would not contain: margin comments from a reviewer (open the Review tab in Word to read them), and three annexes at the end that teach rather than specify.

The companion document is the Technical Design Document (TDD), which converts every requirement in Section 8 into something a developer — or an AI coding agent — can build.

Document Control

Version history

Ver.	Date	Author	Summary of change	Approved by
0.1	29 Jan 2026	K. Menon	Initial draft circulated after discovery workshops (21–24 Jan).	—
0.4	04 Feb 2026	K. Menon	Added integration and data requirements after the SAP walkthrough with R. Balan. Reworked 6 functional requirements that had no testable acceptance criteria.	—
1.0	09 Feb 2026	K. Menon	First baselined version. Signed off by COO and Head of Customer Service.	M. Raghunathan
1.1	18 Feb 2026	K. Menon	CR-01 approved: spare-parts availability check pulled into Phase 1 (adds FR-013, FR-014, INT-003). To protect the go-live date, multilingual technician app (Hindi/Marathi UI) moved to Phase 2 and is now explicitly out of scope — see 4.2. No change to budget.	M. Raghunathan

WHY THIS TABLE EXISTS

Scope changes are the single most common way an FDE engagement goes wrong. Notice what v1.1 actually does: it accepts new scope AND removes equivalent scope in the same breath, in writing, with the sponsor's name against it.

If you take one habit from this document, take that one. 'Yes, and here is what moves out to make room' is the sentence that protects your go-live date.

Distribution list

Name	Role	Organisation
Meera Raghunathan	Chief Operating Officer — Executive Sponsor	Vaayu Pumps & Systems Ltd
Anil Deshpande	Head of Customer Service — Process Owner	Vaayu Pumps & Systems Ltd
Fahad Qureshi	Head of IT Infrastructure & Security	Vaayu Pumps & Systems Ltd
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Sneha Kulkarni	Data Protection Officer	Vaayu Pumps & Systems Ltd
Kabir Menon	Forward Deployed Engineer — Author	AtliQ Technologies
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Related documents

Reference	Document	Status
01-A	Discovery Interview Notes (6 sessions, 21–24 Jan 2026)	Issued
01-C	AS-IS Process Map — Complaint to Closure (swim-lane)	Issued
VPS-FSCC-2026-SDD	Solution Design Document (HLD + LLD)	Draft v0.6

Reference	Document	Status
VPS-FSCC-2026-TDD	Technical Design Document	v1.0 — issued 27 February 2026
VPS-FSCC-2026-UAT	User Acceptance Test Script	Planned — Week 9 of delivery

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1. Executive Summary

Vaayu Pumps & Systems Ltd services an installed base of approximately 11,000 industrial pumps across 1,400 customer sites, supported by 42 field technicians operating from six regional depots. Roughly 180 service complaints arrive each week through three uncontrolled channels — a service mailbox, a WhatsApp number, and the depot phone lines. Every one of them is triaged by hand.

That manual triage now costs Vaayu measurably. Average time from complaint receipt to technician dispatch is 47 minutes. 18% of Priority 1 complaints breach the four-hour contractual response window written into 380 Annual Maintenance Contracts, exposing ₹1.9 crore in annual penalty liability, of which ₹62 lakh was actually paid in FY25. Only 61% of visits achieve a first-time fix; the remaining 39% require a repeat trip, most often because the technician arrived without the correct spare part.

This document specifies an AI-powered Field Service Command Centre that automates intake, triage, dispatch recommendation and spare-parts checking, while keeping a human in the loop for every high-severity or low-confidence decision. The system reads from and writes back to Vaayu's existing SAP S/4HANA Plant Maintenance module. It does not replace SAP PM as the system of record.

The target outcome is a reduction in triage time from 47 minutes to under 5 minutes, a fall in P1 SLA breach rate from 18% to under 6%, and a rise in first-time fix rate from 61% to 75% — worth an estimated ₹40–55 lakh in recovered penalty exposure and approximately 2,100 recovered technician-hours per year.

Delivery is a 14-week engagement with two forward deployed engineers embedded at the Pune plant, with go-live targeted for the Chennai and Pune depots first (Wave 1) before national rollout.

WHAT MAKES THIS AN EXECUTIVE SUMMARY AND NOT AN INTRODUCTION

It answers four questions in order: what is broken, what does it cost, what are we building, what will change. Nothing else.

Note what is absent — no technology names, no architecture, no mention of LLMs or agents. The COO approving this budget does not care, and putting it here signals that you have confused your audience.

Commented [SFR1]: Good executive summary: a COO can read this in 60 seconds and know exactly what they are approving.
Test for yours — if your first paragraph does not contain a number, rewrite it.

2. Business Context

2.1 About the client

Vaayu Pumps & Systems Ltd, established 1994 and headquartered in Pune, manufactures centrifugal pumps and air compressors for the water treatment, pharmaceutical and process industries. FY25 revenue was ₹840 crore. Approximately 34% of gross margin now comes from the after-sales service business rather than equipment sales, which is why service performance has moved onto the board agenda.

Vaayu runs SAP S/4HANA 2022 on-premises, with the PM (Plant Maintenance), MM (Materials Management), SD (Sales & Distribution) and FI/CO modules live. SAP PM holds the equipment master, functional locations, maintenance work orders and service notifications. It is, and will remain, the system of record for all service execution.

2.2 Current state — how a complaint is handled today

Complaints arrive on three channels with no common queue. Three service coordinators at the Pune head office triage all of them manually.

#	Step performed today	Performed by	Typical duration
1	Complaint received by email, WhatsApp message or phone call. Phone calls are transcribed into a shared Excel sheet.	Service coordinator	2–5 min
2	Coordinator searches SAP (transaction IE03) for the equipment, often from a partial serial number or a customer's informal description.	Service coordinator	8–15 min
3	Coordinator reads the complaint and assigns a priority based on personal judgement. There is no written rule set.	Service coordinator	3–6 min
4	Coordinator checks the AMC contract in SD to confirm the customer's SLA tier.	Service coordinator	4–8 min
5	Coordinator telephones two to four technicians in the relevant region to find who is available and appropriately skilled.	Service coordinator	12–25 min
6	Coordinator creates the work order in SAP PM (transaction IW31) and communicates the assignment by phone.	Service coordinator	6–10 min
7	Technician travels to site. Spare-part availability is discovered on arrival, not before.	Field technician	—

The full current-state process, including the escalation and re-visit loops, is documented in the AS-IS swim-lane diagram (reference 01-C).

2.3 Problem statement

Manual triage of field service complaints is slow, inconsistent and unmeasured. It produces four quantified business problems:

Commented [SFR2]: Every one of these numbers came from somewhere. In your own BRD, add the source in brackets — 'ServiceDesk export, Apr–Dec 2025, n=6,412'. A number without a source is an opinion, and the client's finance team will treat it as one.

Problem	Description	Current measure	Annual impact
P-1 Slow triage	47 minutes average from complaint receipt to technician dispatch, of which roughly 60% is lookup and telephone coordination.	47 min average (ServiceDesk export, Apr–Dec 2025, n = 6,412)	≈ 2,100 coordinator-hours
P-2 SLA breaches	P1 complaints under AMC carry a contractual 4-hour response window with penalty clauses. Breaches are detected after the fact, never predicted.	18% P1 breach rate	₹1.9 cr exposure; ₹62 lakh paid FY25
P-3 Repeat visits	Technicians arrive without the correct spare part because stock is not checked before dispatch.	61% first-time fix rate	≈ 1,850 avoidable trips
P-4 Unused telemetry	IoT sensors fitted to 2,300 pumps since 2023 stream to Azure IoT Hub. The data has never been connected to the service process.	0% of work orders telemetry-triggered	Predictive capability forgone

THE 'SO WHAT' TEST

Read each row right to left. If you delete the last column, does anyone still care? If not, the problem does not belong in a BRD.

P-4 is the honest edge case: its impact column is qualitative because nobody can yet price a capability that has never run. It is included because the sponsor asked for it, and it is flagged as Should-have rather than Must-have in Section 8. Say so plainly rather than inventing a number.

3. Business Objectives and Success Metrics

These are the measures against which the engagement will be judged at the post-go-live review. They are deliberately few. Each maps to a problem in Section 2.3.

ID	Objective	Baseline	Target	Measurement method and window
BO-1	Reduce complaint-to-dispatch time	47 min	< 5 min	Median across all complaints auto-triaged without human override. Measured over 30 consecutive days following Wave 2 go-live, Pune and Chennai depots.
BO-2	Reduce P1 SLA breach rate	18%	< 6%	P1 complaints under AMC where first technician response exceeded the contractual window, as a percentage of all P1 AMC complaints, same 30-day window.
BO-3	Improve first-time fix rate	61%	≥ 75%	Work orders closed on the first technician visit as a percentage of all work orders closed, same 30-day window. Excludes orders cancelled by the customer.
BO-4	Reduce coordinator manual effort	≈ 60% of role	≤ 25% of role	Time-and-motion sample of the three service coordinators over 5 working days, run before go-live and repeated 30 days after.
BO-5	Establish an auditable decision trail	None	100%	Every AI-generated priority, dispatch and parts recommendation is retrievable with its inputs, model version and confidence for 24 months. Verified by audit sample of 50 work orders.

Commented [SFR3]: Baseline, target, and how it is measured — all three, or the objective is unfalsifiable. Note the measurement window. 'First-time fix rate 75%' is meaningless without 'measured over the 30 days after Wave 2 go-live, Pune and Chennai depots only'. Clients will hold you to whichever reading suits them if you do not fix it here.

A TRAP TO AVOID

Do not let a client write 'improve customer satisfaction' as a BRD objective unless there is an existing instrument that already produces a number you can move. Vaayu runs no post-service CSAT survey today, so a CSAT target was proposed in discovery and deliberately rejected here — you cannot be held to a baseline that does not exist.

That refusal is itself a piece of FDE craft. Write down what you declined to promise, and why.

4. Scope

4.1 In scope

- Automated intake of service complaints from the Vaayu service mailbox, the WhatsApp Business number and coordinator-entered phone transcripts.
- AI-assisted triage: equipment identification, severity classification, probable-fault diagnosis with citation to the equipment manual, and required technician skill codes.
- Spare-parts availability check against SAP MM depot stock before dispatch (added in v1.1 under CR-01).
- Technician dispatch recommendation based on skill match, geography and shift availability.
- Human-in-the-loop supervisor review queue for all P1 complaints and any recommendation below the confidence threshold.
- Write-back of approved work orders to SAP PM, which remains the system of record.
- SLA clock tracking and predictive breach alerting for AMC customers.
- Supervisor console with review queue, decision audit trail and a correction-capture mechanism.
- Ingestion of IoT telemetry from Azure IoT Hub for the 2,300 instrumented pumps, limited to threshold-based anomaly alerts.
- Operational KPI dashboard covering the five objectives in Section 3.
- End-user training for 3 coordinators, 6 depot supervisors and 42 technicians, and a documentation handover pack.

4.2 Out of scope

The following are explicitly excluded from this engagement. Each exclusion carries a reason and, where relevant, the phase in which it may be reconsidered.

Excluded	Reason for exclusion	Revisit in
Multilingual technician mobile app (Hindi / Marathi UI)	Moved out under CR-01 to make room for the spare-parts check without moving the go-live date. Also requires a Hindi and Marathi evaluation set that does not exist and cannot be built to an acceptable standard before go-live.	Phase 2
Machine-learning predictive failure models on IoT telemetry	Phase 1 delivers threshold-based anomaly alerts only. A trained predictive model requires at least 18 months of labelled failure history, which Vaayu does not currently hold in a usable form.	Phase 2, after data collection
Automatic closure of work orders	Work order closure has financial and warranty consequences and must remain a human action. Not a technical limitation — a deliberate control decision agreed with the Head of Customer Service.	Not planned
Changes to SAP PM configuration, custom Z-tables or ABAP development inside SAP	Vaayu IT policy forbids third-party development inside the SAP core. All integration is read/write through released APIs only.	Not planned
Customer-facing self-service complaint portal	Requested by the sponsor in discovery. Deferred — it changes the intake contract and would double the UAT surface in Phase 1.	Phase 2

Commented [SFR4]: Out-of-scope is the most valuable section in the whole document and the one juniors leave blank. Each exclusion here names a reason. 'Not in this phase' is not a reason. 'Requires a Marathi-language evaluation set we do not have and cannot build before go-live' is a reason.

Excluded	Reason for exclusion	Revisit in
Migration of historical service orders older than 36 months	Retrieval quality gains beyond 36 months were judged marginal against the extraction effort during discovery.	Not planned
Integration with the finance module for automated penalty computation	Depends on FI/CO changes owned by a different internal team with no committed capacity in this window.	Phase 2

4.3 Phasing and rollout

Wave	Scope	Depots	Target
Wave 0	Pilot with 2 coordinators, shadow mode — AI recommends, human always decides, nothing writes to SAP	Pune	Week 9
Wave 1	Live write-back to SAP PM, supervisor review queue active	Pune, Chennai	Week 12
Wave 2	Full feature set including IoT alerts and parts check	Pune, Chennai	Week 14
Wave 3	National rollout — out of scope of this BRD, covered by a separate rollout plan	All six depots	Post-engagement

SHADOW MODE IS NOT OPTIONAL

Wave 0 runs the AI alongside the humans with no ability to act. It exists so that you can measure agreement between the AI's recommendation and the coordinator's decision on real traffic before anything is at stake.

Clients will push to skip it. Do not. It is the cheapest evidence you will ever get, and if agreement is poor it is far better to learn that in Week 9 than in the go-live review.

5. TO-BE Process

The target process replaces steps 1 to 6 of the current state described in Section 2.2. Steps marked [AI] are performed by the system; steps marked [HUMAN] require a person. The full future-state swim-lane is maintained in the Solution Design Document.

#	Actor	Step	Change from today
1	[AI]	Complaint is received on any of the three channels and normalised into a single intake record with a channel-independent schema.	Replaces the shared Excel sheet. One queue instead of three.
2	[AI]	Equipment is identified and matched against the SAP equipment master, including fuzzy match on partial serial numbers and customer-informal descriptions.	Replaces 8–15 minutes of manual IE03 searching.
3	[AI]	Severity is classified P1–P4 against a written rule set, and the AMC SLA tier is resolved from the SD contract.	Replaces personal judgement with an auditable rule set.
4	[AI]	Probable fault is diagnosed by retrieval over the equipment manual and the last 36 months of service history, returned with citations.	New capability. Did not exist.
5	[AI]	Required skill codes and probable spare parts are determined; depot stock is checked in SAP MM.	New capability. Root cause of the 39% repeat-visit rate.
6	[AI]	A dispatch recommendation is produced ranking available technicians by skill match, distance and shift availability, with a stated confidence.	Replaces 12–25 minutes of telephone coordination.
7	[HUMAN]	If severity is P1, or confidence is below threshold, or the parts check failed, the recommendation is routed to the depot supervisor review queue for approval, amendment or rejection.	New control. Nothing high-stakes is automated end to end.
8	[AI]	On approval — or automatically for P3/P4 above the confidence threshold — the work order is created in SAP PM and the technician is notified.	Replaces manual IW31 entry.
9	[AI]	The SLA clock starts. Warning is raised at 80% of the window; escalation fires at 100%.	Breaches are now predicted rather than discovered.
10	[HUMAN]	Supervisor corrections are captured and fed back to improve future recommendations.	New. Allows the system to improve without a developer.

6. Stakeholders and RACI

Stakeholder	Role in this project	Interest	Influence
Meera Raghunathan, COO	Executive sponsor. Owns the business case and the budget.	High	High — decision maker
Anil Deshpande, Head of Customer Service	Process owner. Signs off requirements, owns UAT and adoption.	High	High — day-to-day authority
Fahad Qureshi, Head of IT Infra & Security	Gatekeeper for environments, network access, on-prem deployment and security review.	Medium	High — can block go-live
Ritu Balan, SAP PM Functional Lead	Owns SAP access, API enablement and functional correctness of the write-back.	Medium	High — single point of dependency
Sneha Kulkarni, Data Protection Officer	Reviews DPDP compliance for personal data in complaints and telemetry.	Low	Medium — approval gate
Service coordinators (3)	Primary end users. Their working day changes most.	High	Low formally, high in practice
Depot supervisors (6)	Operate the review queue. Approve or override AI recommendations.	High	Medium
Field technicians (42)	Receive dispatch. Most affected, least consulted by default.	Medium	Low — but adoption depends on them

THE ROW PEOPLE FORGET

Field technicians have low formal influence and are usually not in the steering committee. They are also the group that can quietly kill the project by ignoring the app and continuing to take dispatch by phone.

Two of the six discovery interviews were conducted with technicians in the depot, not in a meeting room. If your stakeholder table only contains people who attend meetings, you have mapped the org chart, not the process.

7. Functional Requirements

Priority is expressed using MoSCoW: **M** = Must have (go-live blocker), **S** = Should have (delivered in Phase 1 if capacity allows, does not block go-live), **C** = Could have.

Every requirement below has an acceptance criterion that can be independently tested. Requirements are traced to design sections in the TDD; the traceability matrix is maintained in that document.

Commented [SFR5]: Read FR-005's acceptance criterion out loud. Someone who has never met you could write a test for it, run it, and tell you pass or fail without asking a single question. That is the whole bar. If a requirement cannot be failed, it is not a requirement — it is a hope.

ID	Name	Requirement	Pri	Acceptance criteria
FR-001	Multi-channel intake	The system shall ingest service complaints from the Vaayu service mailbox (Microsoft 365), the WhatsApp Business number, and phone transcripts entered by a coordinator, and normalise them into a single intake record.	M	A complaint submitted on each of the three channels produces an intake record with identical schema within 60 seconds of receipt. Channel of origin is retained on the record.
FR-002	Structured field extraction	The system shall extract customer identity, site, equipment reference, reported symptom, reported time of failure and reporter contact from unstructured complaint text.	M	Across a labelled test set of 200 historical complaints, each field is extracted with $\geq 92\%$ accuracy. Any field the system cannot extract is returned as null and flagged, never guessed.
FR-003	Equipment identification	The system shall match the complaint to a specific equipment record in the SAP PM equipment master, including where the customer supplies a partial serial number, an informal description or a tag number.	M	$\geq 88\%$ of the 200-complaint test set match to the correct equipment number. Where confidence is below 0.80 the system returns the top three candidates for human selection rather than choosing one.
FR-004	Duplicate detection	The system shall detect when an incoming complaint refers to an equipment and fault already covered by an open work order, and link rather than duplicate it.	S	A second complaint for the same equipment number with a semantically similar symptom raised within 72 hours of an open order is linked to that order and does not create a new one. Coordinator can unlink.
FR-005	Severity classification	The system shall assign a severity of P1 to P4 to every complaint using the published severity rule set in Annex A of the TDD.	M	Severity is assigned within 5 seconds of intake completion. Agreement with the coordinator's decision on the Wave 0 shadow-mode sample is $\geq 90\%$ for P1 and $\geq 80\%$ overall. Every assignment records which rule fired.
FR-006	Probable fault diagnosis	The system shall propose up to three probable causes for the reported symptom, drawn from the equipment manual corpus and the last 36 months of closed service orders for the same equipment model.	M	Every proposed cause carries a citation to a manual page or a closed service order number. A cause with no retrievable source is not shown. Where retrieval returns nothing above the relevance threshold, the system states that no probable cause was found.
FR-007	Skill code determination	The system shall determine the technician skill codes required to attend the complaint, based on equipment model and probable fault.	M	Skill codes returned are valid entries in the Vaayu technician skill matrix. Where the probable fault is unresolved, the system returns the baseline skill code for that equipment class.
FR-008	Spare parts prediction	The system shall predict the spare parts likely required, expressed as SAP material numbers with a likelihood band of High, Medium or Low.	S	Predicted parts are valid SAP material numbers. On the historical test set, the part actually consumed appears in the High or Medium band in $\geq 70\%$ of cases.
FR-009	Confidence threshold and human routing	The system shall compute a confidence score for every triage outcome and route to the supervisor review queue where confidence is below the configured threshold,	M	Threshold is configurable by an administrator without a code change or redeployment. No P1 complaint is ever dispatched without human approval, verified across the full UAT set. Routing reason is recorded on the record.

ID	Name	Requirement	Pri	Acceptance criteria
		where severity is P1, or where any upstream integration failed.		
FR-010	Technician matching	The system shall rank available technicians for a complaint by skill match, road distance from the functional location, and remaining shift capacity.	M	Only technicians holding at least one required skill code and on shift within the SLA window are returned. Ranking inputs and their weights are visible to the supervisor on the recommendation.
FR-011	Dispatch recommendation with rationale	The system shall present its dispatch recommendation together with a plain-language rationale a supervisor can read in under 15 seconds.	M	Rationale states the selected technician, the skill match, the distance, and the reason the next-ranked candidate was not selected. Assessed by 6 supervisors during UAT; ≥ 5 of 6 must rate it comprehensible without explanation.
FR-012	Supervisor override	The system shall allow a supervisor to accept, amend or reject any recommendation, and shall require a reason code on rejection.	M	Override is possible on every field of the recommendation. Rejection without a reason code is not accepted. Original recommendation is retained alongside the override, never overwritten.
FR-013	Depot stock availability check	The system shall check unrestricted stock for each predicted spare part at the depot serving the complaint site before a dispatch recommendation is issued. (Added v1.1, CR-01)	M	Stock position is read from SAP MM and is no more than 15 minutes old at the point of recommendation. Parts in the High likelihood band that are unavailable are shown to the supervisor with the nearest depot holding stock.
FR-014	Parts shortfall handling	Where a High-likelihood part is unavailable at the serving depot, the system shall route the recommendation to the supervisor queue regardless of confidence, and propose the nearest alternative depot or a stock transfer. (Added v1.1, CR-01)	M	No auto-dispatch occurs when a High-likelihood part is unavailable. The alternative proposal names a depot, a distance and a stock quantity, or states explicitly that no alternative exists within 150 km.
FR-015	SLA tier resolution and clock start	The system shall resolve the applicable SLA tier from the customer's AMC contract in SAP SD and start the response clock at the timestamp of complaint receipt, not at the time of triage.	M	SLA tier matches the contract record for 100% of AMC customers in the UAT set. Clock start timestamp equals channel receipt timestamp. Non-AMC customers are handled under the default internal target and labelled as such.
FR-016	SLA breach prediction and escalation	The system shall raise a warning at 80% of the elapsed response window and escalate at 100%, to the depot supervisor and the Head of Customer Service.	M	Warning and escalation fire within 60 seconds of the threshold. Escalation recipients are configurable per depot. Escalations are recorded and reportable.
FR-017	SAP PM work order creation	On approval, the system shall create a maintenance work order in SAP PM carrying equipment number, functional location, priority, assigned technician, required skill codes and predicted parts.	M	Work order is visible in SAP PM (IW33) within 30 seconds of approval with all fields populated. Failure to write produces a retry and, if still failing, an alert to the coordinator — never a silent failure.
FR-018	Decision audit trail	The system shall record, for every AI-generated decision, the inputs, retrieved sources, model and prompt version, confidence, output and any human override.	M	Any work order created in the last 24 months can be reconstructed to show why the system recommended what it did. Verified by audit sample of 50 work orders during UAT.
FR-019	Supervisor console	The system shall provide a web console showing the review queue, each recommendation with its rationale and confidence, and the SLA clock for every open item.	M	Queue is sorted by SLA time remaining ascending by default. A supervisor can action an item in no more than three clicks from the queue view.
FR-020	Correction capture and reuse	The system shall capture supervisor corrections and reason codes, and	S	A correction of the same class recurring three times is surfaced in the weekly review report.

ID	Name	Requirement	Pri	Acceptance criteria
		apply them to improve subsequent recommendations without a code deployment.		Corrections are retrievable as a labelled dataset for evaluation. No model retraining is in scope for Phase 1.

Note on FR-009. The confidence threshold, the P1 rule and the integration-failure rule are three independent routes into the human review queue. They are deliberately not combined into a single condition, so that any one of them can be tuned or disabled without affecting the others.

Note on FR-020. Phase 1 captures corrections and makes them available; it does not retrain a model. This distinction is written into the acceptance criterion precisely because clients hear 'the system learns' and assume something more than is being built.

Commented [SFR6]: FR-009 is the most important requirement in this document, and it is a Must-have for a reason.

An enterprise AI system without a confidence threshold and a route to a human is not a product, it is a liability. Get this into your own BRDs by default.

Commented [SFR7]: FR-020 is what Principle 8 of the roadmap (Always Include a Memory Layer) looks like when it is written as a requirement rather than a good intention.

8. Integration Requirements

SAP S/4HANA remains the system of record. All access is through released SAP interfaces. No development inside the SAP core is in scope (see Section 4.2).

ID	Interface	Direction	Method and auth	Volume / frequency	Owner
INT-001	SAP PM — Equipment Master and Functional Location	Read	OData v2, API_EQUIPMENT and API_FUNCLOCATION via SAP Gateway. OAuth 2.0 client credentials.	Full sync nightly; on-demand lookup per complaint, ≈ 200/day	R. Balan (Vaayu SAP)
INT-002	SAP PM — Maintenance Order create and update	Write	BAPI_ALM_ORDER_MAINTAIN exposed through SAP Integration Suite (CPI). OAuth 2.0, mTLS on the CPI channel.	≈ 180/week, peak 40/day	R. Balan (Vaayu SAP)
INT-003	SAP MM — Material stock at depot (added v1.1, CR-01)	Read	OData API_MATERIAL_STOCK_SRV, filtered by storage location. OAuth 2.0 client credentials.	≈ 3 calls per complaint, cached 15 min	R. Balan (Vaayu SAP)
INT-004	SAP SD — AMC service contract header and SLA tier	Read	OData API_SERVICE_CONTRACT. OAuth 2.0 client credentials.	1 call per complaint, cached 24 h	R. Balan (Vaayu SAP)
INT-005	Microsoft 365 service mailbox	Read	Microsoft Graph API, application permissions, Azure AD app registration.	Poll every 60 s	F. Qureshi (Vaayu IT)
INT-006	WhatsApp Business Platform	Read / Write	Cloud API webhook inbound; template messages outbound for technician notification.	≈ 60 inbound/day	A. Deshpande (Vaayu CS)
INT-007	Azure IoT Hub — pump telemetry	Read	Event Hub compatible endpoint, consumer group dedicated to this system. Managed identity.	2,300 devices, 1 reading / 5 min	F. Qureshi (Vaayu IT)
INT-008	Equipment manual corpus	Read (one-time + delta)	SFTP drop of PDF manuals from the Vaayu document management system.	≈ 340 manuals, quarterly delta	R. Balan (Vaayu SAP)

Commented [SFR8]: This is the section that makes or breaks an enterprise AI project, and the one that most junior BRDs reduce to a single line reading 'integrate with SAP'. Name the interface, the direction, the authentication, the volume and the owner. The owner column is not decoration — INT-002 has a named human against it because it is the dependency most likely to slip.

WHAT A GOOD FDE ASKS FOR IN THE FIRST WEEK

You cannot write this table from a conversation. You get it by asking the client's IT team for four specific things: the data dictionary, the API catalogue with which services are actually enabled in their landscape, the authorisation objects your service user will need, and sandbox access.

Ask on day one. INT-002 in this project took eleven working days to get enabled, and that number was known and planned for because it was asked for early. This is the scarce skill the roadmap refers to in Week 9 — not the ability to call an OData endpoint, but knowing what to demand and when.

9. Data Requirements

ID	Data set	Requirement	Notes
DR-001	Equipment master	Complete extract of active equipment for the 1,400 in-scope customer sites, refreshed nightly.	≈ 11,000 records. Known issue: 6% carry a null or malformed serial number — see RSK-03.
DR-002	Historical service orders	36 months of closed maintenance orders including long text, fault codes and consumed materials.	≈ 21,000 orders. Used for retrieval in FR-006 and for the evaluation set.
DR-003	Equipment manual corpus	PDF service and maintenance manuals for all in-scope equipment models.	≈ 340 documents. Approximately 40 are scanned images and require OCR.
DR-004	Technician skill matrix	Current skill codes, certifications with expiry, home depot and shift roster for all 42 technicians.	Held in a spreadsheet today, not in SAP HR. Ownership and refresh process must be agreed — see DEP-04.
DR-005	AMC contract terms	SLA response and resolution windows by contract, and the penalty basis.	380 contracts. Four distinct SLA tiers confirmed in discovery.
DR-006	IoT telemetry	Vibration, bearing temperature and discharge pressure at 5-minute intervals for 2,300 instrumented pumps.	Rolling 90-day retention in the operational store; older data stays in IoT Hub archive.
DR-007	Personal data handling	Complaint records contain reporter name, phone number and email. These are personal data under the DPDP Act 2023.	Retention 24 months, then pseudonymised. Approved by S. Kulkarni, DPO, 06 Feb 2026. No personal data leaves the India region.

WRITE DOWN THE DATA QUALITY PROBLEMS BEFORE YOU SIGN

DR-001 states plainly that 6% of equipment serial numbers are broken. That sentence is worth more to you than any architecture diagram.

If you discover it after go-live, it is your bug. If it is in the BRD the client approved, it is a known constraint with a mitigation — and FR-003's acceptance criterion of 88% rather than 99% is defensible because of it. Data quality is not a technical detail, it is a commercial position.

10. Non-Functional Requirements

ID	Category	Requirement	Verification
NFR-01	Latency	End-to-end triage — intake received to recommendation presented — completes within 90 seconds at P95 under normal load.	Load test at 3× peak volume before Wave 1.
NFR-02	Availability	99.5% during business hours (07:00–22:00 IST, Mon–Sat), measured monthly. Outside those hours, best effort.	Uptime monitoring; monthly report.
NFR-03	Throughput	Sustains 40 complaints per hour with no queue growth; degrades gracefully by queueing, never by dropping.	Load test with Locust at 120 complaints/hour.
NFR-04	Data residency	All personal data and all model inference remain within the India region. No complaint content is sent to a model endpoint hosted outside India.	Architecture review by F. Qureshi and S. Kulkarni before Wave 1.
NFR-05	Deployment	Application runs on Vaayu's on-premises Kubernetes cluster. Only the LLM inference call egresses, to Azure OpenAI in Central India.	Security review sign-off.
NFR-06	Security	No credential in source or environment variable; all secrets in Azure Key Vault. Role-based access control on the supervisor console. All traffic TLS 1.2 or above.	Penetration test before Wave 1; OWASP Top 10 for LLM review.
NFR-07	Prompt injection resistance	Complaint text is untrusted input. The system must not execute instructions embedded in a complaint, and must not disclose system prompts or other customers' data.	Adversarial test set of 40 injection attempts; zero successful.
NFR-08	Auditability	Decision records retained 24 months and exportable to CSV for audit.	Audit sample of 50 records during UAT.
NFR-09	Cost	Model inference cost not to exceed ₹28 per complaint at steady state, monitored daily with an alert at 80% of the monthly budget.	Cost dashboard reviewed weekly from Wave 0.
NFR-10	Model fallback	If the primary model endpoint is unavailable, the system falls back to the secondary endpoint. If both fail, complaints queue for manual triage and coordinators are alerted within 2 minutes.	Failover drill before Wave 1.
NFR-11	Usability	A service coordinator can complete a full review-and-approve cycle after 45 minutes of training with no reference to documentation.	Observed during UAT with all 3 coordinators.

NFR-07 IS THE ONE JUNIORS OMIT

In an enterprise AI system the untrusted input is not a form field, it is the free text a customer wrote. Anyone who emails your service mailbox can put words in front of your model.

Writing it as a numbered, testable requirement is what turns 'we should think about prompt injection' into something UAT can actually fail you on.

11. Assumptions, Dependencies and Constraints

11.1 Assumptions

ID	Assumption	If untrue
ASM-01	SAP S/4HANA API services listed in Section 8 can be enabled in the Vaayu landscape without a support pack upgrade.	Adds 3–4 weeks; INT-002 write-back would need an interim file-based mechanism.
ASM-02	The three service coordinators remain available for 4 hours per week for discovery, UAT and training throughout the engagement.	Requirements and UAT quality degrade; timeline extends.
ASM-03	Vaayu's on-premises Kubernetes cluster has spare capacity of 12 vCPU and 48 GB RAM for this workload.	Additional hardware procurement, 6–8 week lead time.
ASM-04	Equipment manuals are available as text-extractable PDFs, apart from the ~40 known scanned documents.	OCR effort increases; FR-006 citation quality falls.
ASM-05	Historical service order long text is in English or a mix of English and transliterated Hindi, as observed in the discovery sample.	Retrieval quality falls; a translation step would be needed.

11.2 Dependencies

ID	Dependency	Owner	Needed by
DEP-01	SAP sandbox and QA system access for the AtliQ team, with the required authorisation objects.	R. Balan	Week 1
DEP-02	Enablement of the OData services in Section 8 in the Vaayu landscape.	R. Balan	Week 3
DEP-03	Azure OpenAI Central India subscription and quota provisioned under Vaayu's tenant.	F. Qureshi	Week 2
DEP-04	Technician skill matrix confirmed as authoritative, with a named owner and a refresh process.	A. Deshpande	Week 4
DEP-05	DPDP review completed and documented.	S. Kulkarni	Week 5
DEP-06	Six depot supervisors released for 2 days of UAT.	A. Deshpande	Week 11

11.3 Constraints

- No custom development, Z-tables or ABAP inside the SAP core (Vaayu IT policy).
- Personal data may not leave the India region (DPDP Act 2023 and Vaayu internal policy).
- Go-live must complete before 31 May 2026; the June to September monsoon period is the peak service season and no production change is permitted during it.
- Total engagement budget is fixed. Scope added must be traded against scope removed — as demonstrated by CR-01 in the version history.

12. Risks and Mitigation

ID	Risk	Lik.	Imp.	Mitigation and owner
RSK-01	SAP API enablement (DEP-02) slips, blocking all integration work.	M	H	Raised as a named dependency with a Week 3 date and a weekly check-in with R. Balan. Contingency: build against the SAP Business Accelerator Hub sandbox and swap endpoints at integration. Owner: K. Menon.
RSK-02	Coordinators resist the system, seeing it as a threat to their roles.	M	H	Coordinators are positioned as supervisors of the AI, not replaced by it, and are named as reviewers in FR-012. Two of the three were interviewed in discovery. ADKAR-based change plan from Week 8. Owner: A. Deshpande.
RSK-03	Poor equipment master data quality (6% malformed serials) degrades FR-003 match rate below the accepted 88%.	H	M	Acknowledged in DR-001 and reflected in the FR-003 acceptance criterion. Low-confidence matches return three candidates for human selection rather than guessing. Data cleansing proposed as a separate Vaayu-owned activity. Owner: R. Balan.
RSK-04	Model cost exceeds the NFR-09 ceiling at production volume.	M	M	Bottom-up model routing: a lightweight model for classification and extraction, the premium model only for diagnosis. Daily cost tracking from Wave 0 with an 80% alert. Owner: I. Rao.
RSK-05	Prompt injection through complaint text reaches the model.	M	H	NFR-07 with a 40-case adversarial test set. Input sanitisation, strict tool boundaries, and no tool is capable of writing to SAP without human approval on P1. Owner: I. Rao.
RSK-06	Scope creep during UAT as supervisors see the system for the first time.	H	M	Section 4.2 is explicit and signed. Every UAT observation is triaged as defect, clarification or change request, and change requests follow the CR process shown in the version history. Owner: R. Vaidya.
RSK-07	Key-person dependency on R. Balan as the single SAP contact.	M	H	Escalated to the sponsor in Week 2. A named backup and documented API runbook requested as part of DEP-01. Owner: R. Vaidya.
RSK-08	Go-live pushes into the monsoon peak-season freeze.	L	H	Wave plan front-loads write-back to Week 12, leaving two weeks of buffer before the 31 May constraint. Reviewed at every fortnightly steering. Owner: R. Vaidya.

EVERY RISK HAS A NAME AGAINST IT

A risk register where every mitigation owner is 'the project team' is decoration. Notice that some owners here are client-side, not vendor-side — RSK-03 is Vaayu's data problem and the BRD says so.

Assigning a client-side owner in a signed document is not passing the buck. It is how you avoid being blamed in month four for a data quality issue you flagged in month one.

13. Acceptance and Sign-off

This document is baselined at version 1.1. Changes after this point follow the change control process: a written change request, an impact assessment covering scope, timeline and cost, and sponsor approval before any work begins. Approved changes are recorded in the version history in the Document Control section.

Acceptance of this BRD confirms that the signatories agree the problem, the objectives and their measurement, the scope boundary including exclusions, and the requirements in Sections 7 to 10 as the basis for design and build.

Name	Role	Signature	Date
Meera Raghunathan	Executive Sponsor, Vaayu		
Anil Deshpande	Process Owner, Vaayu		
Fahad Qureshi	IT and Security, Vaayu		
Rohan Vaidya	Delivery Manager, AtliQ Technologies		

Annex A — Glossary

Term	Meaning
AMC	Annual Maintenance Contract. A service agreement carrying defined SLA response and resolution windows, often with financial penalty clauses.
BAPI	Business Application Programming Interface. SAP's released, remotely callable function modules — the supported way to write into SAP from outside.
Equipment master	The SAP PM record identifying an individual physical asset, its model, serial number, installation site and maintenance history.
First-time fix rate	Percentage of work orders resolved on the first technician visit, with no repeat trip required.
Functional location	The SAP PM record describing where an asset is installed, independent of which physical asset currently occupies that position.
Human in the loop	A design pattern where the system produces a recommendation but a person makes the decision, for cases that are high-stakes or low-confidence.
IW31 / IW33	SAP transaction codes for creating and displaying a maintenance order.
Master data vs transactional data	Master data describes things that persist (equipment, customers, materials). Transactional data records events (work orders, goods issues). AI systems usually read both and write only transactional.
MoSCoW	Prioritisation scheme: Must have, Should have, Could have, Won't have.
OData	A REST-based protocol SAP uses to expose data to external systems; the usual read path for an FDE.
P1 to P4	Severity bands. P1 is a total production stoppage; P4 is a cosmetic or advisory issue.
Shadow mode	Running a new system alongside the existing manual process with no ability to act, purely to measure agreement before cutover.
System of record	The single authoritative source for a piece of data. Here, SAP PM. An AI layer that disagrees with the system of record is wrong by definition.

Annex B — Requirements That Can Actually Be Tested

This annex is not part of the specification. It exists because the difference between a BRD that works and one that causes a fight in month four is almost entirely in how requirements are worded.

Each row below shows a requirement as it was first written during discovery, why it fails, and the version that was baselined. Every one of these is a real rewrite from the drafting of this document.

As first written (rejected)	Why it fails	As baselined
The system should assign work orders quickly and efficiently.	Neither 'quickly' nor 'efficiently' can be failed. Two people reading this will build different systems and both will believe they complied.	FR-010 / NFR-01: end-to-end triage completes within 90 seconds at P95 under normal load, verified by load test at 3× peak volume.
The AI should be accurate.	Accurate at what, measured on what data, against whose judgement, and how accurate? This is a wish, not a requirement.	FR-002: each extracted field achieves ≥ 92% accuracy across a labelled test set of 200 historical complaints. Unextractable fields return null and are flagged.
The system will integrate with SAP.	SAP is not a thing you integrate with; it is dozens of interfaces with different protocols, owners and enablement states. This sentence hides the entire risk of the project.	Section 8, INT-001 to INT-004: each names the interface, direction, protocol, authentication, volume and a client-side owner with a date.
The system should handle errors gracefully.	Untestable, and worse, it lets a silent failure count as graceful. Silent failure is the worst outcome in an enterprise workflow.	FR-017: failure to write to SAP produces a retry and, if still failing, an alert to the coordinator — never a silent failure. NFR-10 specifies the model fallback chain explicitly.
The system should learn from user feedback.	Clients hear 'learns' and picture a model that retrains itself. If that is not what you are building, this sentence is a promise you will be held to.	FR-020: corrections are captured, surfaced in the weekly review report and retrievable as a labelled dataset. The acceptance criterion states explicitly that no model retraining is in scope for Phase 1.
Priority will be assigned appropriately based on the situation.	'Appropriately' delegates the hardest decision in the project back to the reader. It also makes the requirement impossible to test.	FR-005: severity assigned against the published rule set in Annex A of the TDD, with ≥ 90% agreement with coordinators on P1 in shadow mode, and a record of which rule fired.
The solution must be secure and compliant.	True of everything and actionable for nothing. Security review will not know what to test and the DPO will not know what to approve.	NFR-04, NFR-06, NFR-07, DR-007: named region for inference, named secret store, an adversarial injection test set with a zero-success bar, a stated retention period, and a dated DPO approval.

THE ONE-SENTENCE TEST

Before you baseline any requirement, ask: could a tester who has never met me write a test for this, run it, and tell me pass or fail without asking me a question?

If the answer is no, you have not written a requirement. You have written a note to yourself.

Annex C — Reviewer's Rubric

Use this to grade your own BRD before anyone else reads it. It is the checklist used to review this document at v0.4, which is why v0.4 required rework on six requirements.

Score each line Yes or No. Any No in the Blocking column means the document is not ready to circulate.

#	Check	Blocking	Where in this document
1	Does the executive summary contain at least one quantified business impact within the first paragraph?	Yes	Section 1
2	Does every problem statement carry a current measure and a source for that measure?	Yes	Section 2.3
3	Does every objective have a baseline, a target and a stated measurement window?	Yes	Section 3
4	Is there an explicit out-of-scope section, and does each exclusion give a reason rather than just a phase?	Yes	Section 4.2
5	Could a tester write a pass/fail test for every functional requirement without asking the author a question?	Yes	Section 7, Annex B
6	Does every requirement carry a priority, and is the Must-have set genuinely the go-live blocking set?	Yes	Section 7
7	Is there a defined route to a human for high-stakes or low-confidence AI decisions?	Yes	FR-009, FR-012
8	Does every integration name the interface, direction, protocol, authentication, volume and a client-side owner?	Yes	Section 8
9	Are known data quality problems written down rather than discovered later?	Yes	DR-001, RSK-03
10	Are non-functional requirements testable, including latency, security and cost?	Yes	Section 10
11	Is prompt injection addressed as a numbered requirement, not a paragraph of intent?	Yes	NFR-07
12	Does every risk have a named individual as mitigation owner, including client-side owners where the risk is client-side?	Yes	Section 12
13	Is there a version history showing what changed and who approved it?	Yes	Document Control
14	When scope was added, was equivalent scope removed or the timeline or cost formally revised?	Yes	v1.1 / CR-01
15	Is there a glossary, so a new joiner or an auditor can read the document unaided?	No	Annex A
16	Does the document avoid naming technologies in the business sections?	No	Sections 1–4

Annex D — How This Document Feeds the Next One

A BRD is not a deliverable in isolation. It is one link in a chain, and its quality is measured by how little the next document has to guess.

This BRD section	Feeds	Becomes, in the TDD
Section 7 — Functional Requirements	Component and agent design	Agent responsibilities, tool boundaries, and the traceability matrix mapping every FR to a design section
FR-002, FR-003 — extraction and matching	Field-level validation specification	Mandatory fields, formats, fuzzy-match thresholds and null-handling rules
FR-005, FR-015, FR-016 — severity and SLA	Business rule specification	The severity decision table and the SLA escalation rule set, written so a developer implements them without interpretation
FR-009 — confidence and routing	Control flow and state design	Routing conditions, threshold configuration and the review queue state machine
Section 8 — Integration Requirements	Integration specification	Endpoint, payload contract, auth flow, retry and idempotency policy, and error codes per interface
Section 9 — Data Requirements	Data model	Entities, relationships, retention and the pseudonymisation rule for personal data
Section 10 — Non-functional Requirements	Observability, security and cost design	Trace fields, metrics, alert thresholds, model routing policy and the fallback chain
Section 7 acceptance criteria	UAT test script	Test scenarios with pre-conditions, steps and expected results — written from the acceptance criteria, not invented afresh

WHY THIS MATTERS MORE THAN IT LOOKS

If a requirement in Section 7 has a vague acceptance criterion, that vagueness does not stay in the BRD. It propagates into the TDD as a design gap, into the code as an assumption, and into UAT as an argument about what was agreed.

The reverse is also true, and it is the practical reason to write acceptance criteria well: a good criterion is already 80% of a UAT test case. Write it once, properly, and Week 23 gets much easier.